Christopher J Cox

christopher.j.cox@noaa.gov

Curriculum Vitae

November 2, 2026

Google Scholar • GitHub • LinkedIn

Education

Ph.D. Environmental Science University of Idaho, Moscow (2013)
M.S. Geography, Certificate GIS University of Idaho, Moscow (2009)
B.A. Anthropology University of Maine, Orono (2006)

Appointments

Physical Scientist IV, Team Lead National Oceanic and Atmospheric Administration (NOAA) Physical Sciences Laboratory (2022-)
Physical Scientist III National Oceanic and Atmospheric Administration (NOAA) Physical Sciences Laboratory (2019-2022)
Research Scientist II CIRES/University of Colorado (2017-2019)
Research Scientist I CIRES/University of Colorado (2014-2017)
Postdoctoral Visting Fellow CIRES/University of Colorado (2013-2014)
Research Assistant Department of Geography, University of Colorado (2007-2013)

Skills

Research: meteorology, snow, sea ice, albedo, surface energy budget, radiation, turbulence, cloud physics, model validation, climate
Hardware: instrumentation/integration, uncrewed aerial systems, analog & digital data acquisition
Software: MATLAB, Python, CRBASIC, git, Linux/Unix
Methods: radiative transfer modeling (e.g., LBLRTM, DISORT, RRTM, Streamer, SFIT), statistical modeling (e.g., linear inverse), thermo/fluid dynamics (e.g., bulk aerodynamics, diffusion), wavelet analysis, gridded data products, weather observing
Core Competencies: complex problem solving; project leadership & management; technical writing; collaboration & interdisciplinary work; adaptability & continuous learning; environmental data analytics; scientific programming
Project Management: Principal Investigator NOAA/GOMO Arctic Research Program, 2019-present

Awards & Honors

NOAA Bronze Medal for Scientific or Engineering Achievement (2021, 2022)
University of Colorado Postdoctoral Visiting Fellowship (2013)
University of Idaho Outstanding Doctoral Research and Creativity Award (2013)

Service (selected)

NOAA/OAR Uncrewed Systems (UxS) Executive Oversight Board (2026-)
University-National Oceanographic Laboratory System (UNOLS) Arctic Icebreaker Coordination Committee (2022-)
United Nations Decade of Ocean Science (UNDOS) Safe Ocean Action Plan & Observing Air-Sea Interaction Strategy Working Groups
Baseline Surface Radiation Network (BSRN) Cold Climates Issues Working Group Chair (2016-2021)
Mentor: CIRES Mentoring Program; NOAA Lapenta Internship; NOAA Hollings Scholar Program; NOAA/UCAR SOARS

Fieldwork

More than 1.5 years of fieldwork since 2010, inclusive of 9 months arctic winter, 1.5 months equatorial Pacific, 1 month western USA: ice sheet, ocean, land, sea ice, mountain snowpack. FAA Part 107 Certified. Past trainings include: tower climber; wilderness first responder; arctic field training; unescorted oil field access; marine standards of training and certification of watchkeeping.

Peer-Reviewed Publications

Total: > 70 • Publications > 100 citations: 6 • H-Index: 28 • i10-Index: 58 • Selected:

Cox, C.J. et al. (2026) Resiliency of Arctic sea ice to warming from southerly advection in early spring. Elementa, 14(1), 00027, https://doi.org/10.1525/elementa.2026.00027

Cox, C.J. and C. Penland (2026) Medium range predictability of the wintertime Bering Sea ice edge using linear inverse modeling. *Journal of Geophysical Research - Oceans*, 131(3), e2025JC023413, https://doi.org/10.1029/2025JC023413

Hossain, A., …, **C.J. Cox**, et al. (2026) Machine learning eliminates reanalysis warm bias and reveals weaker winter surface cooling over Arctic sea ice. *Geophysical Research Letters*, 53(12), e2025GL121289, https://doi.org/10.1029/2025GL121289

Schwat, E., … **C.J. Cox**, et al. (2025) Estimating snow sublimation in complex terrain: A season of intensive field measurements and the role of vertical water vapor divergence. *Journal of Hydrometeorology*, 26(10), 1455-1473, https://doi.org/10.1175/JHM-D-25-0022.1

Sledd, A., M.D. Shupe, A. Solomon, and **C.J. Cox** (2025) Surface energy balance responses to radiative forcing in the central Arctic from MOSAiC and models. *Journal of Geophysical Research – Atmospheres*, 130(6), e2024JD042578, https://doi.org/10.1029/2024JD042578

Cummins, D.P., V. Guemas, M.R. Gallagher, **C.J. Cox**, and M.D. Shupe (2023) Surface turbulent fluxes from MOSAiC campaign predicted by machine learning. *Geophysical Research Letters*, 50(23), e2023GL105698, https://doi.org/10.1029/2023GL105698

Cox, C.J. et al. (2022) Continuous observations of the surface energy budget and meteorology over Arctic sea ice during MOSAiC. *Scientific Data*, 10, 519, https://doi.org/10.1038/s41597-023-02415-5

de Boer, G., S. Borenstein, R. Calmer, **C. Cox**, M. Rhodes, C. Choate, J. Hamilton, J. Osborn, D. Lawrence, B. Argrow, and J. Intrieri (2022) Measurements from the University of Colorado RAAVEN Uncrewed Aircraft System during ATOMIC. *Earth System Science Data*, 14, 19-31, https://doi.org/10.5194/essd-14-19-2022

Cox, C.J. et al. (2021) The De-Icing Comparison Experiment: A study of broadband radiometric measurement under icing conditions in the Arctic. *Atmospheric Measurement Techniques*, 14, 1205-1224, https://doi.org/10.5194/amt-14-1205-2021

Cox, C.J. et al. (2019) The Aleutian Low – Beaufort Sea Anticyclone: A climate index correlated with seasonal melt in the Pacific Arctic cryosphere. *Geophysical Research Letters*, 46, GRL59183, https://doi.org/10.1029/2019GL083306

Solomon, A., G. de Boer, J. Creamean, A. McComiskey, M. Shupe, M. Maahn, and **C.J. Cox** (2018) The relative impact of cloud condensation nuclei and ice nucleating particle concentrations on phase-partitioning in Arctic Mixed-Phase Stratocumulus Clouds. *Atmospheric Chemistry and Physics*, 18, 17047-17059, https://doi.org/10.5194/acp-18-17047-2018

Rowe, P.M., S.P. Neshyba, **C.J. Cox**, and V.P. Walden, (2016): Towards autonomous surface-based infrared remote sensing of polar clouds: Cloud height retrievals. *Atmospheric Measurement Techniques*, 9, 3641-3659, https://doi.org/10.5194/amt-9-3641-2016

Berkelhammer, M., D. Noone, H.C. Steen-Larson, M. O'Neill, A. Bailey, **C.J. Cox**, D. Schneider, K. Steffen, and J.C. White, (2016) Surface-atmosphere decoupling limits accumulation over Greenland. *Science Advances*, 2, e1501704, https://doi.org/10.1126/sciadv.1501704

Cox, C.J., V.P. Walden, P.M. Rowe, and M.D. Shupe (2015) Humidity trends imply increased sensitivity to clouds in a warming Arctic. *Nature Communications*, 6, 1-8, https://doi.org/10.1038/ncomms10117

Cox, C.J., V.P. Walden, G.P. Compo, P.M. Rowe, M.D. Shupe, and K. Steffen (2014) Wavelet analysis of downwelling longwave flux and cloud radiative forcing from surface observations and ERA-Interim over Summit, Greenland. *Journal of Geophysical Research*, 119(21), 12317-12337, https://doi.org/10.1002/2014JD021975

Bennartz, R., M.D. Shupe, D.D. Turner, V.P. Walden, K. Steffen, **C.J. Cox**, M.S. Kulie, N.B. Miller, C. Pettersen (2013) July 2012 Greenland melt extent enhanced by low-level liquid clouds. *Nature*, 496, 83-86. https://doi.org/10.1038/nature12002

Fieldwork Photography

















